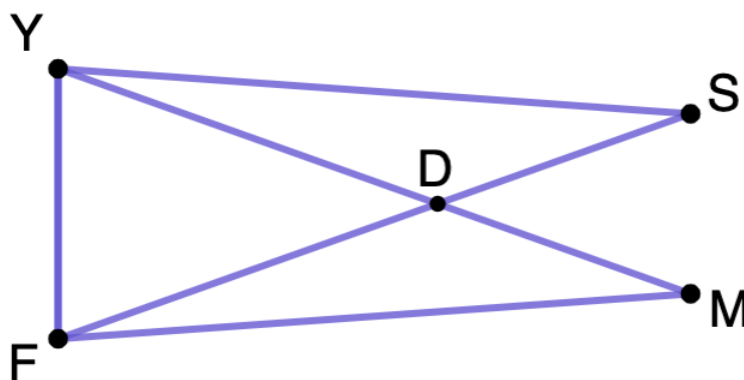


5.8.2 Activity

Proof A

Given: $\angle YSD \cong \angle FMD$
 $\triangle YDF$ is isosceles
 with base $\overline{YF}$

Prove: $\overline{FM} \cong \overline{YS}$

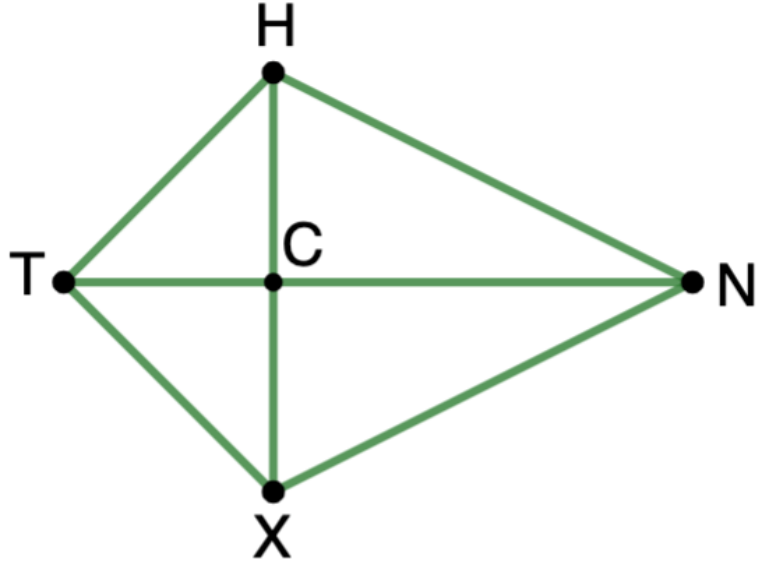


Statement	Reason
$\angle YSD \cong \angle FMD$	Given
$\triangle YDF$ is isosceles with base $\overline{YF}$	Given
$\overline{YD} \cong \overline{FD}$	Definition of isosceles triangle
$\angle SDY \cong \angle MDF$	Vertical angle theorem
$\triangle DMF \cong \triangle DSF$	AAS congruence theorem
$\overline{FM} \cong \overline{YS}$	CPCTC

Proof B

Given: $\overline{TN}$ bisects $\angle HTX$
 $\overline{HT} \cong \overline{XT}$

Prove: $\overline{TN}$ bisects $\angle HNX$

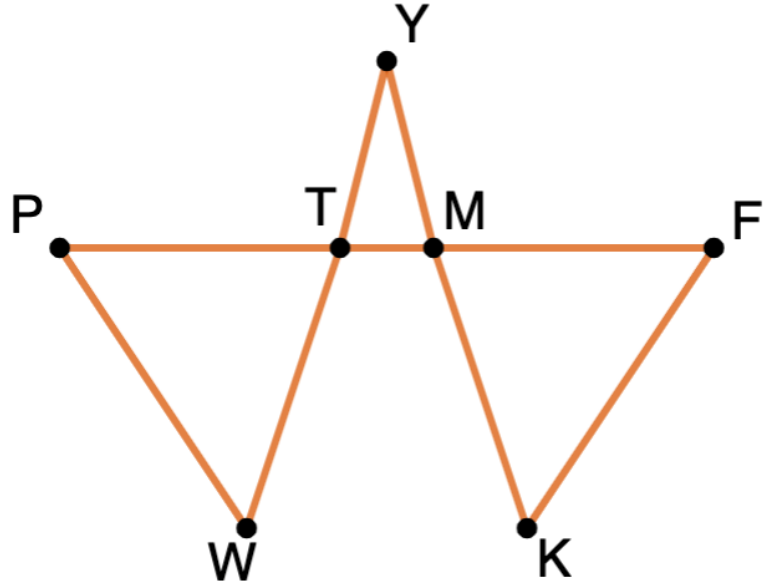


Statement	Reason
$\overline{HT} \cong \overline{XT}$	Given
$\overline{TN}$ bisects $\angle HTX$	Given
$m\angle HTC = m\angle XTC$	Definition of angle bisector
$\angle HTC \cong \angle XTC$	Definition of congruence
$\overline{TN} \cong \overline{TN}$	Reflexive property of congruence
$\triangle TXN \cong \triangle THN$	SAS congruence postulate
$\angle HNT \cong \angle XNT$	CPCTC
$m\angle HNT = m\angle XNT$	Definition of congruence
$\overline{TN}$ bisects $\angle HNX$	Definition of angle bisector

Proof C

Given: $\angle WPT \cong \angle KFM$
 $\overline{PW} \cong \overline{FK}$
 $\angle YMT \cong \angle YTM$

Prove: $\triangle PWT \cong \triangle FKM$

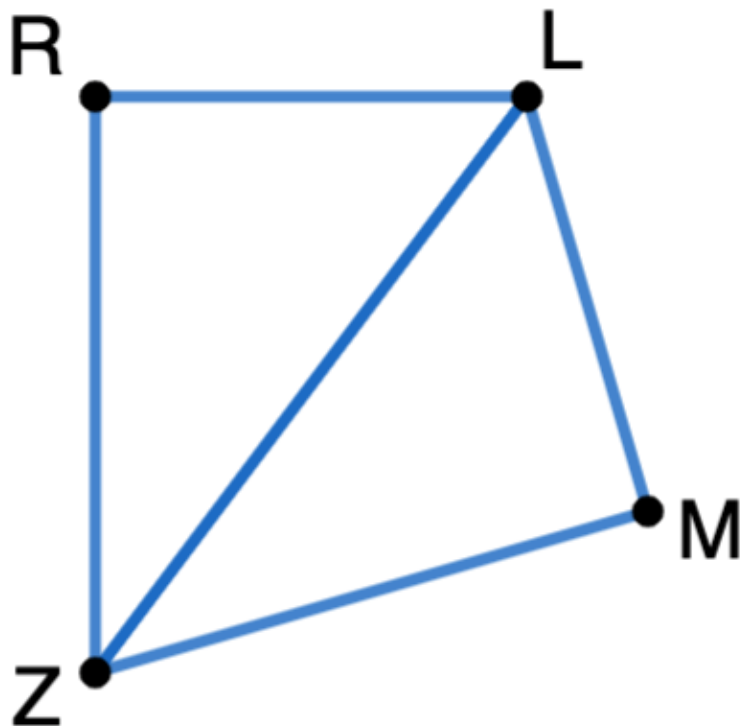


Statement	Reason
$\angle WPT \cong \angle KFM$	Given
$\overline{PW} \cong \overline{FK}$	Given
$\angle YMT \cong \angle YTM$	Given
$\angle YMT \cong \angle FMK$	Vertical angles theorem
$\angle YTM \cong \angle PTW$	Vertical angles theorem
$\angle FMK \cong \angle PTW$	Substitution property
$\triangle PWT \cong \triangle FKM$	AAS congruence postulate

Proof D

Given: $\overline{LM} \cong \overline{LR}$
 $\overline{LR} \perp \overline{RZ}$
 $\overline{LM} \perp \overline{MZ}$

Prove: $\angle MZL \cong \angle RZL$

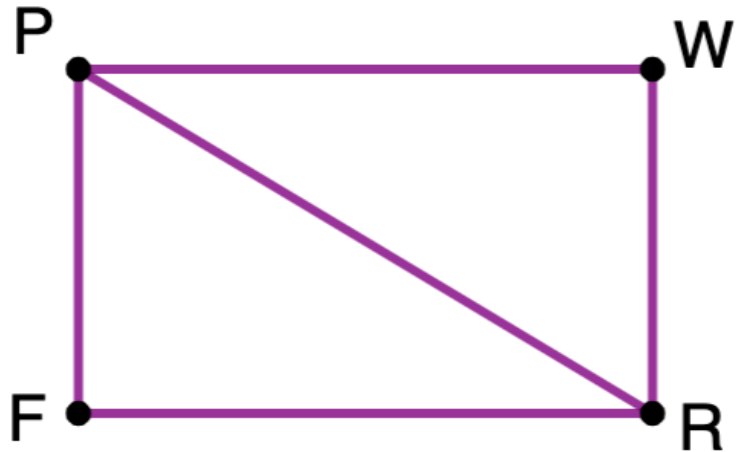


Statement	Reason
$\overline{LR} \perp \overline{RZ}, \overline{LM} \perp \overline{MZ}$	Given
$\angle LMZ$ and $\angle LRZ$ are right angles	Definition of perpendicular lines
$\triangle LMZ$ and $\triangle LRZ$ are right triangles	Definition of right triangles
$\overline{LM} \cong \overline{LR}$	Given
$\overline{LZ} \cong \overline{LZ}$	Reflexive property of congruence
$\triangle LMZ \cong \triangle LRZ$	HL theorem
$\angle MZL \cong \angle RZL$	CPCTC

Proof E

Given: $\overline{PF} \cong \overline{WR}$
 $\overline{PW} \cong \overline{FR}$

Prove: $\overline{PF} \parallel \overline{WR}$

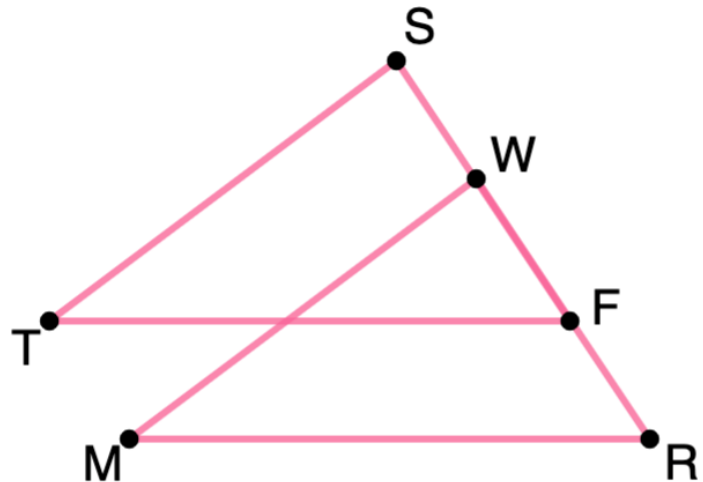


Statement	Reason
$\overline{PF} \cong \overline{WR}$	Given
$\overline{PW} \cong \overline{FR}$	Given
$\overline{PR} \cong \overline{PR}$	Reflexive property of congruence
$\triangle PRF \cong \triangle RPW$	SSS congruence postulate
$\angle FPR \cong \angle WRP$	CPCTC
$\overline{PF} \parallel \overline{WR}$	Converse of alternate interior angles theorem

Proof F

Given: $\overline{RF} \cong \overline{SW}$
 $\angle TSF \cong \angle MWR$
 $\overline{FT} \parallel \overline{MR}$

Prove: $\triangle TSF \cong \triangle MWR$



Statement	Reason
$\overline{RF} \cong \overline{SW}$	Given
$RF = SW$	Definition of congruence
$FW = FW$	Reflexive property of equality
$RF + FW = SW + FW$	Addition property of equality
$RF + FW = RW$ and $SW + FW = SF$	Segment addition postulate
$RW = SF$	Substitution property
$\overline{RW} \cong \overline{SF}$	Definition of congruence
$\overline{FT} \parallel \overline{MR}$	Given
$\angle TFS \cong \angle MRW$	Corresponding angle theorem
$\angle TSF \cong \angle MWR$	Given
$\triangle TSF \cong \triangle MWR$	ASA congruence theorem